TA Assignment

Edge Cases & Solutions

Pre-Assignment + Threshold-Based Activation Strategy

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Update Notice:

This document introduces the pre-assignment concept and covers 25 edge cases not addressed in the original TA_System_Implementation_Plan.pdf. 10 spec gaps are identified.

1. CONTEXT & CURRENT ARCHITECTURE

The TA system is built on course-level assignment that auto-propagates to all sessions. This document analyzes edge cases for a proposed enhancement: pre-assigning TAs at course creation and activating them automatically when NELC enrollment thresholds are met.

Current Architecture

- TA assignment is course-level (not session-level), auto-propagates to all sessions
- Manager/Admin assigns directly (status='approved'), Instructor can only suggest (status='suggested')
- NELC thresholds: <=25 students = 0 TAs | 26-35 = 1 TA | >35 = 2 TAs
- Auto-allocation is DISABLED per client request - system shows warnings only
- course_assistants table tracks: status (suggested/approved/rejected), suggestedBy, approvedBy

Proposed Enhancement: Pre-Assignment vs Activation

Term	Definition
Pre-assigned	TA is linked to a course with status 'pre_assigned'. They receive an email notification but are NOT propagated to sessions yet. They appear with a 'Waiting' badge.
Activated	TA status changes to 'approved'. They are propagated to all sessions, can view attendance/submissions, and join Zoom. Triggered automatically when enrollment crosses the NELC threshold.
Deactivated	TA status changes back to 'pre_assigned' (not deleted). They are removed from future sessions. Triggered when enrollment drops below the threshold.

New Status Flow

```
Manager assigns at course creation
        |
  [pre_assigned] ---- enrollment crosses threshold ----> [approved/activated]
        ^                                                        |
        '---- enrollment drops below threshold -----------------'

Instructor suggests
        |
  [suggested] -> Manager approves -> [pre_assigned] -> threshold met -> [approved]
              -> Manager rejects  -> [rejected]
```

Schema Changes Required

- Add new status value 'pre_assigned' to course_assistants.status CHECK constraint
- Add new column 'priority' (integer, default 1) to determine activation order
- Add new column 'deactivation_scheduled_at' (nullable timestamp) for grace period

2. EDGE CASES

EC-1: Pre-Assignment at Course Creation

Scenario:

Course Manager creates a course and assigns a TA at the same time (before any students enroll).

Problem:

With 0 students, NELC requires 0 TAs. Should the TA be activated immediately?

Solution:

- TA is stored in course_assistants with status = 'pre_assigned' and priority = 1
- TA receives a pre-assignment notification email: 'You have been pre-assigned to [Course Name]. You will be activated when enrollment reaches the required threshold.'
- TA sees the course on their dashboard with a 'Pre-assigned (Waiting)' badge but cannot access sessions, attendance, or submissions yet
- TA is NOT propagated to any sessions at this point
- Manager UI shows the TA with a yellow 'Pre-assigned' badge: 'Will activate when enrollment reaches 26 students'

EC-2: Assigning Two TAs at Course Creation

Scenario:

Course Manager pre-assigns TA1 and TA2 to a course at creation time (0 students enrolled).

Problem:

Should both be assigned? In what order should they activate?

Solution:

- Both TAs are stored with status = 'pre_assigned'
- TA1 gets priority = 1, TA2 gets priority = 2 (based on assignment order)
- When enrollment hits 26: only TA1 (priority=1) activates
- When enrollment hits 36: TA2 (priority=2) also activates
- Both TAs receive pre-assignment emails at course creation
- Manager can reorder priority via drag-and-drop or up/down arrows in the TA management panel
- UI shows: 'TA1 - Priority 1 (activates at 26 students) | TA2 - Priority 2 (activates at 36 students)'

EC-3: Threshold Crossed Upward - First TA Needed

Scenario:

Course has 25 students (0 TAs required). Student #26 enrolls. Now 1 TA is required.

Problem:

Which TA gets activated? What if no TA is pre-assigned?

Solution:

- If a pre-assigned TA exists: The TA with priority = 1 is automatically activated
- Status changes from 'pre_assigned' to 'approved'
- TA is propagated to ALL existing sessions via propagateToAllSessions()
- TA receives activation email: 'You have been activated for [Course Name]'
- Instructor receives in-app notification: 'TA [name] has been activated for your course'
- NELC audit log: ASSISTANT_ALLOCATED
- If NO pre-assigned TA exists: System sends warning to all Managers, NELC scorecard marks TA category as 'failed', Instructor sees red warning badge

EC-4: Threshold Crossed Upward - Second TA Needed

Scenario:

Course has 35 students with 1 active TA (TA1). Student #36 enrolls. Now 2 TAs are required.

Problem:

Where does TA2 come from?

Solution:

- If TA2 is pre-assigned (priority=2): Automatically activated (same flow as EC-3)
- If only TA1 was pre-assigned (no TA2): System sends warning to all Managers, NELC scorecard shows '1 of 2 required TAs assigned' (partial compliance)
- If TA2 was suggested by instructor but still pending approval: System sends urgent notification to Managers to approve or assign

EC-5: Threshold Crossed Downward - Drop from 2 TAs to 1 TA

Scenario:

Course has 36 students with 2 active TAs (TA1 priority=1, TA2 priority=2). A student unenrolls -> 35 students. Now only 1 TA is required.

Problem:

Which TA gets deactivated? What happens to their session data?

Solution:

- The TA with the highest priority number (lowest priority) gets deactivated: TA2 (priority=2) is deactivated, TA1 (priority=1) stays active
- TA2 status changes from 'approved' to 'pre_assigned'
- TA2 is removed from all future/upcoming sessions (WHERE session status = 'scheduled')
- TA2 is KEPT on completed/ended sessions (historical data preserved for NELC audit)
- TA2 receives notification about deactivation, Instructor gets in-app notification
- NELC audit log: ASSISTANT_DEALLOCATED
- 24-hour grace period: Do not deactivate immediately. If enrollment goes back above threshold within 24h, cancel the deactivation. Prevents flapping.

EC-6: Threshold Crossed Downward - Drop from 1 TA to 0 TAs

Scenario:

Course has 26 students with 1 active TA. Student unenrolls -> 25 students. Now 0 TAs required.

Problem:

Should the TA be fully removed?

Solution:

- TA status changes from 'approved' to 'pre_assigned' (NOT deleted)
- Same deactivation flow as EC-5 (remove from future sessions, keep on completed)
- TA stays as pre-assigned so they can be reactivated if enrollment goes back up
- Manager can explicitly remove the TA entirely if desired (DELETE endpoint)
- Same 24-hour grace period applies

EC-7: TA Pre-Assigned to Multiple Courses - One Activates

Scenario:

TA-X is pre-assigned to Course A and Course B. Course A reaches 26 students first -> TA-X is activated for Course A. Course B still has 20 students.

Problem:

Does activating TA-X for Course A affect Course B?

Solution:

- No impact on Course B. Each course has its own independent course_assistants row
- TA-X is now: Course A = 'approved' (active), Course B = 'pre_assigned' (waiting)
- A TA can be active in multiple courses simultaneously - by design (like university TAs)
- TA dashboard shows both courses with clear status: Course A 'Active' (green), Course B 'Pre-assigned' (yellow)

EC-8: TA Pre-Assigned to Multiple Courses - Both Activate Simultaneously

Scenario:

TA-X is pre-assigned to Course A (24 students) and Course B (25 students). Two students enroll simultaneously - Course A hits 26, Course B hits 26.

Problem:

Can a TA be activated in two courses at the same time?

Solution:

- Yes. A TA can be active in multiple courses simultaneously
- Each enrollment event triggers its own threshold check independently
- Both activations happen in their own database transactions
- TA-X receives two activation emails (one per course)
- No conflict - UNIQUE constraint is on (courseId, userId), same user can exist in multiple courses
- Scheduling conflict consideration: Show a warning (not blocker) if courses have overlapping session times

EC-9: TA Workload Limit

Scenario:

TA-X is already active in 5 courses. Manager tries to pre-assign TA-X to a 6th course.

Problem:

Should there be a maximum number of courses a TA can be assigned to?

Solution:

- System should allow it (no hard limit) but show a workload warning to Manager
- Warning: 'TA [name] is currently active in 5 courses and pre-assigned to 2 more'
- Manager makes the final decision (per NELC: assignment is institution-controlled)
- Optional future enhancement: Configurable max course limit per TA in system settings

EC-10: Instructor Suggests TA Already in Another Course

Scenario:

Instructor of Course B suggests TA-X, who is already pre-assigned/active in Course A.

Problem:

Should the system prevent this? Should Managers know about existing assignments?

Solution:

- Allow the suggestion - do not block it
- Suggestion notification to Managers includes: 'Note: TA [name] is currently assigned to [N] other course(s)'
- Manager can make an informed decision to approve or reject
- If approved, TA-X is pre-assigned to Course B independently

EC-11: Student Enrollment Fluctuates Around Threshold (Flapping)

Scenario:

Course has 25-26 students going back and forth as students enroll/unenroll repeatedly. TA keeps getting activated/deactivated.

Problem:

Constant activate/deactivate is disruptive and generates excessive notifications/emails.

Solution:

- 24-hour grace period for deactivation: When enrollment drops below threshold, a deactivation timer starts
- If enrollment goes back above within 24h, the timer is cancelled - no deactivation
- If 24h passes and enrollment is still below, deactivation executes
- Upward activation is immediate - no delay when students enroll (TA is needed right away)
- Rate limiting on notifications: Maximum 1 threshold notification per course per 24 hours
- Implementation: Add 'deactivation_scheduled_at' column + hourly cron job

EC-12: Manager Removes Pre-Assigned TA Before Activation

Scenario:

Manager pre-assigned TA1 and TA2 at course creation. Before any threshold is reached, Manager decides to remove TA1.

Problem:

What happens to the priority order?

Solution:

- TA1 is deleted from course_assistants
- TA2's priority is automatically promoted from 2 to 1
- If enrollment later reaches 26, TA2 (now priority=1) activates
- NELC audit log: ASSISTANT_DEALLOCATED for TA1

EC-13: Manager Replaces an Active TA

Scenario:

TA1 is active (approved) in a course with 30 students. Manager wants to replace TA1 with TA3.

Problem:

How to do a clean swap without breaking NELC compliance?

Solution:

- Two-step process (recommended): 1) Manager assigns TA3 (approved immediately), 2) Manager removes TA1
- OR one-step 'Replace' action in the UI: assign TA3 + remove TA1 in a single transaction
- At no point does the course have 0 TAs (compliance maintained)
- TA1 gets removal notification, TA3 gets assignment email

EC-14: Course Instructor Changes Mid-Course

Scenario:

Course is reassigned to a different instructor. The original instructor had suggested TA-X (status='suggested', pending manager approval).

Problem:

What happens to pending TA suggestions?

Solution:

- Pending suggestions (status='suggested') are kept but flagged
- New instructor sees: 'TA [name] was suggested by [Previous Instructor] - Pending manager approval'
- New instructor can leave it for manager or suggest a different TA
- Pre-assigned and active TAs are not affected by instructor changes (course-level, not instructor-level)

EC-15: TA Account Deactivated/Suspended

Scenario:

TA-X is active in Course A (30 students, 1 TA required). Admin deactivates TA-X's user account.

Problem:

Course now has 0 TAs when 1 is required. NELC non-compliant.

Solution:

- When user account is deactivated: all course_assistants entries set to 'rejected', removed from future sessions
- NELC audit log: ASSISTANT_DEALLOCATED
- Immediate warning to all Managers: 'TA [name] account deactivated. Courses need replacement.'
- If a priority-2 TA is pre-assigned, they are automatically promoted and activated
- NELC scorecard updates immediately to reflect the deficit

EC-16: Course Deletion or Archival

Scenario:

Manager archives/deletes a course that has active and pre-assigned TAs.

Problem:

What happens to TA assignments?

Solution:

- Course archived: TA assignments preserved in DB (for NELC audit trail), TAs no longer see course on active dashboard
- Course deleted (CASCADE): course_assistants rows deleted via CASCADE DELETE on courseId FK
- NELC audit log retains historical records independently
- TAs receive notification: 'Course [name] has been archived'

EC-17: Sessions Exist Before Any TA Is Pre-Assigned

Scenario:

Instructor creates 5 sessions in a course. Later, Manager pre-assigns a TA. Then enrollment hits 26.

Problem:

When the TA activates, do they get propagated to the 5 existing sessions?

Solution:

- Yes. When TA status changes from 'pre_assigned' to 'approved', propagateToAllSessions() runs
- Queries ALL sessions for the course and inserts TA into session_assistants for each
- Uses onConflictDoNothing (safe for re-runs)
- Same flow that already exists for assignTA() - activation path reuses it

EC-18: Enrollment Import / Bulk Enrollment

Scenario:

Manager imports a CSV of 50 students into a course that currently has 0 students. This should trigger TA activation for 2 TAs simultaneously.

Problem:

Does the threshold check happen per-student or at the end of the batch?

Solution:

- Batch-aware threshold check: After bulk enrollment completes, run ONE threshold evaluation (not 50 individual checks)
- Evaluation sees: 0 -> 50 students, requires 2 TAs
- If TA1 (priority=1) and TA2 (priority=2) are pre-assigned: both activate
- If only TA1 is pre-assigned: TA1 activates + warning for missing second TA
- Implementation: Enrollment service calls evaluateThresholds(courseId) once after batch completes

EC-19: Race Condition - Concurrent Enrollments

Scenario:

Two students enroll at the exact same moment (two API requests). Course has 25 students. Both requests see 25 and try to process enrollment #26.

Problem:

Could the threshold activation trigger twice?

Solution:

- Threshold evaluation must be idempotent - calling it twice with 26 students activates TA only once
- propagateToAllSessions() already uses onConflictDoNothing (duplicate insertions safe)
- Status update uses optimistic lock: UPDATE ... WHERE status = 'pre_assigned' (only one request succeeds)
- If both requests try to activate, one succeeds and one is a no-op

EC-20: Mixed Course-Level and Manual Session-Level Assignment

Scenario:

TA-X is pre-assigned to Course A at course level. Meanwhile, Manager manually assigns TA-X to a specific session via session-assistants endpoint.

Problem:

When the course-level TA activates, it tries to propagate to sessions - but TA-X is already on that session.

Solution:

- onConflictDoNothing on UNIQUE constraint (session_id, user_id) handles this gracefully
- Manual session assignment has isAutoAssigned = false, which is preserved (not overwritten)
- When TA-X is deactivated at course level, only isAutoAssigned = true rows are removed
- Manually assigned sessions remain (respects the isAutoAssigned flag)

EC-21: Threshold Changes (System Configuration Update)

Scenario:

Admin changes NELC thresholds from (25/35) to (20/30). Several courses now fall above the new thresholds.

Problem:

Should existing courses retroactively activate/deactivate TAs?

Solution:

- Threshold changes should trigger a full re-evaluation of all active courses
- Background job iterates all active courses: recalculate requiredTAs with new thresholds
- If deficit: activate pre-assigned TAs (if available) or send warnings
- If surplus: schedule deactivation (with 24h grace period)
- Managers receive summary notification with count of affected courses
- NELC audit log records the threshold change event

EC-22: TA Reactivation After Deactivation

Scenario:

TA-X was deactivated from Course A (enrollment dropped 26 -> 25). Two days later, enrollment goes back to 26.

Problem:

Should the same TA be reactivated, or should the system pick a different one?

Solution:

- Same TA is reactivated - they retain their 'pre_assigned' status and priority from the deactivation
- Reactivation flow is identical to initial activation (EC-3)
- TA-X receives a reactivation email
- Propagation to sessions runs again (safe due to onConflictDoNothing)

- Historical session access from previous active period was already preserved

EC-23: No Available TAs in the System

Scenario:

NELC threshold says 1 TA is needed, but there are zero users with TEACHING_ASSISTANT role in the system.

Problem:

The system can't satisfy the requirement.

Solution:

- NELC scorecard marks TA category as 'failed'
- Warning sent to all Managers and Admins: 'No Teaching Assistants available'
- Admin dashboard shows banner: 'TA deficit across [N] courses. Consider creating new TA accounts.'
- System does NOT auto-assign random users or promote students - strictly institution-controlled (per NELC)

EC-24: TA Pre-Assigned via Instructor Suggestion (Full Flow)

Scenario:

Instructor suggests TA-X for Course B. Manager approves. Course B has 20 students.

Problem:

After manager approval, should the TA be activated immediately or pre-assigned?

Solution:

- Manager approval changes status from 'suggested' to 'pre_assigned' (NOT directly to 'approved')
- TA-X receives pre-assignment notification, activates only when enrollment reaches 26
- Exception: If enrollment is already at or above threshold at time of approval, TA is activated immediately
- approveTA() method checks current enrollment: if >= threshold -> 'approved' + propagate, else -> 'pre_assigned'

EC-25: Multiple Managers Act on Same TA Simultaneously

Scenario:

Manager A approves TA-X's suggestion while Manager B rejects it at the same time.

Problem:

Race condition on status update.

Solution:

- Use optimistic concurrency: UPDATE ... WHERE status = 'suggested'
- First request to execute wins; second gets 'TA suggestion has already been processed'
- Frontend shows toast with info about who already processed it

3. SUMMARY TABLE

#	Edge Case	Trigger	Solution Summary
1	Pre-assign at creation	0 students	Status = pre_assigned, email, waiting badge
2	Two TAs at creation	0 students	Both pre_assigned with priority 1 and 2
3	25 -> 26 students	Enrollment up	Priority-1 TA auto-activates, or warning
4	35 -> 36 students	Enrollment up	Priority-2 TA auto-activates, or warning
5	36 -> 35 students	Enrollment down	Priority-2 deactivated (24h grace)
6	26 -> 25 students	Enrollment down	TA to pre_assigned (24h grace)
7	TA in 2 courses, 1 activates	Independent	No cross-course impact
8	Both courses activate	Simultaneous	Yes, TA active in multiple courses
9	TA overloaded	Workload	Warning to Manager, no hard limit
10	Suggest TA from other course	Suggestion	Allow, show assignments to Manager
11	Enrollment fluctuation	Flapping	24h deactivation grace period
12	Remove pre-assigned TA	Manager action	Priority auto-reorders
13	Replace active TA	Manager action	Assign new first, then remove old
14	Instructor changes	Reassignment	Suggestions kept, TAs unaffected
15	TA account deactivated	Admin action	Warn Managers, promote backup TA
16	Course archived/deleted	Manager action	Preserve audit trail, notify TAs
17	Sessions before TA	Late assignment	propagateToAllSessions() retroactive
18	Bulk enrollment (CSV)	0 -> 50 students	Single threshold check after batch
19	Concurrent enrollments	Race condition	Idempotent, optimistic locking
20	Mixed course + session	Manual override	onConflictDoNothing, isAutoAssigned flag
21	Threshold config change	Admin settings	Background re-evaluation of all courses
22	Reactivation	Enrollment recovers	Same TA reactivates, retains priority
23	No TAs in system	Empty pool	NELC fails, warning to Admins
24	Suggestion + approval	Below threshold	approved if met, else pre_assigned
25	Concurrent managers	Race condition	Optimistic concurrency, first-wins

4. GAPS IN CURRENT PDF SPEC

The following edge cases are NOT covered in the existing TA_System_Implementation_Plan.pdf:

1. No pre-assignment concept - the PDF only has 'suggested' / 'approved' / 'rejected'. There is no 'waiting for threshold' state.
2. No priority ordering - the PDF doesn't define which TA activates first when multiple are assigned.
3. No deactivation on enrollment drop - the PDF only covers assignment, not what happens when students leave.
4. No grace period - the PDF doesn't address enrollment fluctuation (flapping).
5. No workload visibility - no mention of tracking how many courses a TA is assigned to.
6. No bulk enrollment handling - no guidance on threshold behavior during CSV imports.
7. No TA account deactivation cascade - no coverage of what happens when a TA's account is disabled.
8. No session scheduling conflict detection - no mention of warning when a TA's courses have overlapping sessions.
9. No threshold configuration change handling - thresholds are hardcoded with no re-evaluation strategy.
10. Auto-allocation is disabled - the PDF says 'stays DISABLED' but doesn't provide the pre-assignment alternative that bridges the gap.

5. RECOMMENDED IMPLEMENTATION PRIORITY

Priority	Edge Cases	Reason
P0	EC-1, EC-2, EC-3, EC-4, EC-5, EC-6, EC-7, EC-8, EC-17, EC-19	Core pre-assignment + activation flow
P1	EC-11, EC-12, EC-13, EC-15, EC-18, EC-24	Stability, compliance, common operations
P2	EC-9, EC-10, EC-14, EC-16, EC-20, EC-21, EC-22, EC-23, EC-25	Edge scenarios, admin tooling

This document was prepared by Gamma Developers for the PhysioGuidance eLearning Platform. All implementation must comply with NELC Health Sector eLearning Standards, SCFHS/CPD requirements, WCAG 2.1 AA accessibility, and Saudi PDP Data Protection Law.